

Lab Report for Power Electronics

Lab 1: Switching devices

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1 INTRODUCTION

1.1 Research Question/ Problem Statement

The objective of this lab is to develop a better understanding of basic electronic switching technologies in AC and DC circuits. This will be investigated through the use of a thyristor, TRIAC, and MOSFET and IGBT circuits. First we compare a thyristor versus a TRIAC in an AC circuit. Then, we analyse the differences between MOSFETs and IGBTs in a DC circuit. Finally, we take a look at clamping circuits.

1.2 Background/ Theory

The current measurement is done by a sense resistor. Using Ohms law ($I = \Delta V/R$) we can determine the current.

An inductive load will react to a change in current. The induced voltage is equal to, $V = -L \frac{di}{dt}$, where $\frac{di}{dt}$ is the rate of change in current.

2 METHOD & MATERIALS

2.1 Experimental Set-up

The studied circuits are those provided in the lab text. The reader is invited to reference this text to see all circuits or can consult the corresponding section in the lab text

2.2 Measuring Instruments

This lab took place in the lab 10.03. The instruments used and their specifications are listed below.

- Aim TTi power supply
- Rhode and Schwarz HMO 1002
- Tektronix signal generator AFG 1022

Since the specifications of these devices are less relevant for these measurements we refer to the datasheets of these devices.

2.3 Method

By measuring voltage and current and relate this to the input signals, the behavior of the circuit can be determined and an explanation for the behavior can be formulated.

3 RESULTS THYRISTOR VERSUS TRIAC

3.1 Thyristor

3.1.1 Measurements

The first circuit is an AC dimmer circuit with a Thyristor. A Thyristor is a three-terminal electrical component that conducts current in one direction when it is triggered on its gate. It keeps conducting until the current goes below a certain threshold. The thyristor circuit is shown in Figure 1 below. The diode ensures that there is no negative current going to the gate. The RC circuit with a potentiometer introduces a phase shift, altering the point within the cycle at which the thyristor is triggered.

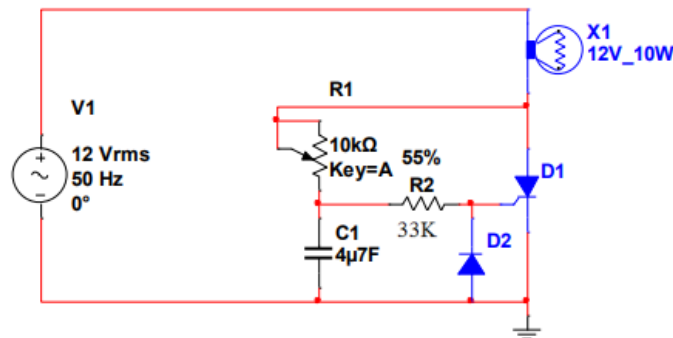
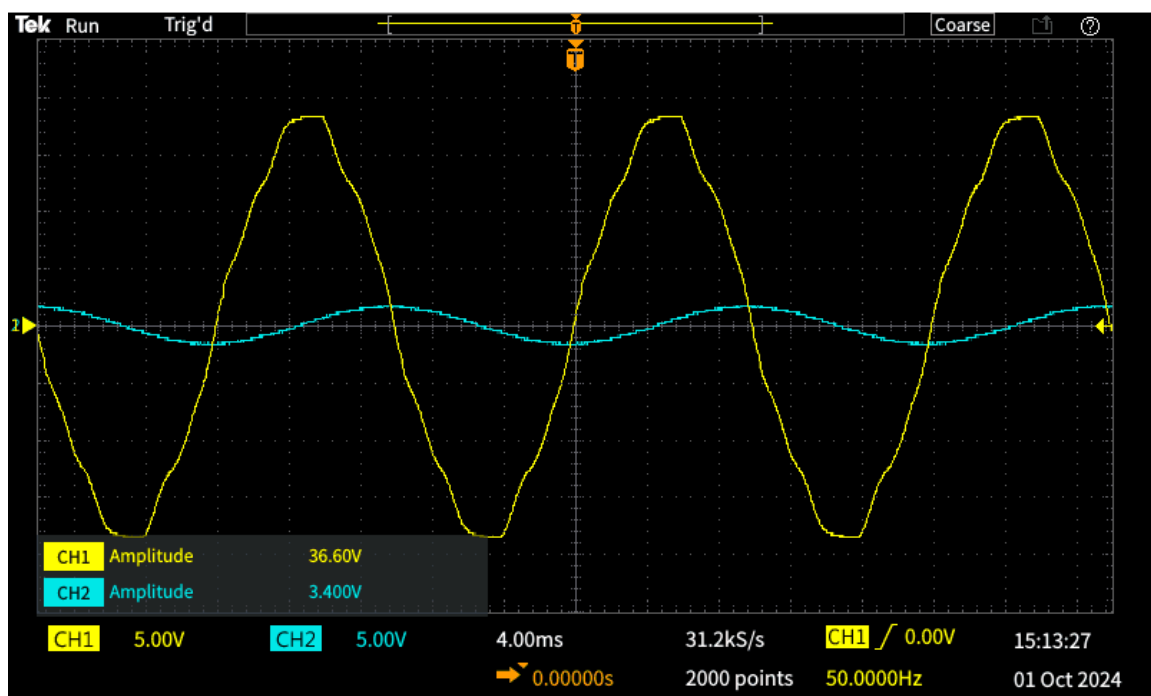


Figure 1 Thyristor AC dimmer circuit

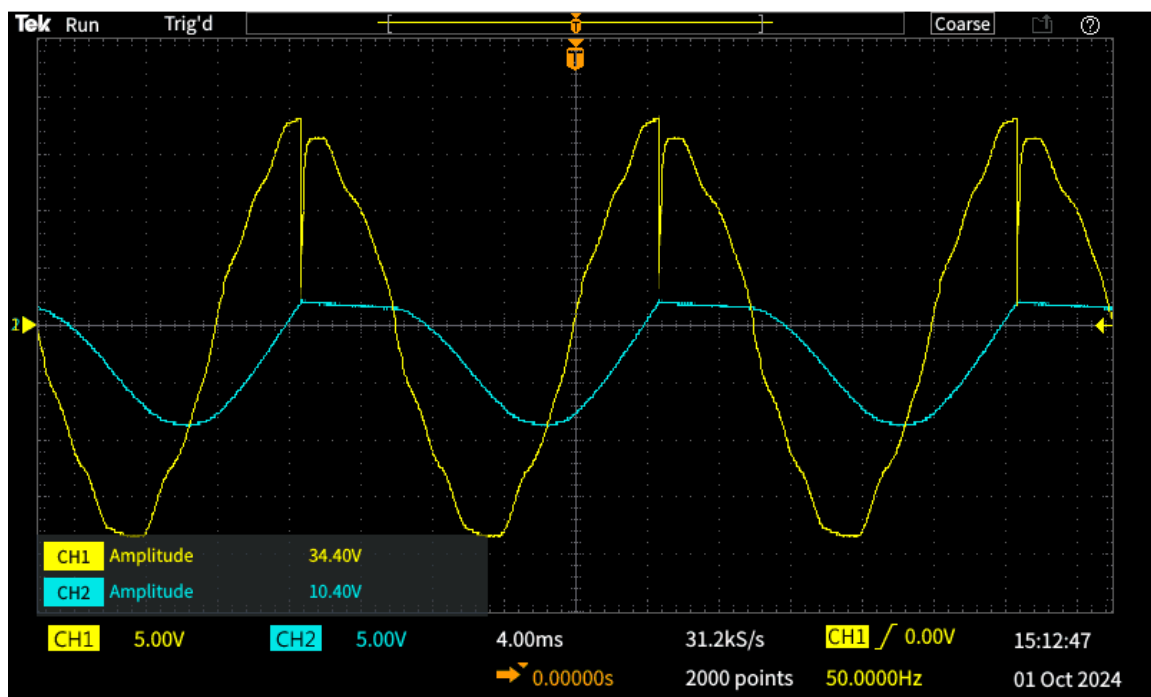
3.1.2 Graphs

The following figures show the voltage over the thyristor with the potentiometer at 0%, 50%, and 100% respectively. First, we show the input signal and signal at the gate of the thyristor. Then, we subtract the input signal from the signal over the thyristor to determine the current thru the load. On the scope images we can clearly see the phase shift when turning the potentiometer and when it activates when the gate signal is at a certain voltage. On the last 3 scope images we can see the current is flowing when the thyristor is activated. The voltage drop on the input is due to the lamp being cold at this moment where it's almost a short circuit. When the lamp gets hotter the resistance increases.



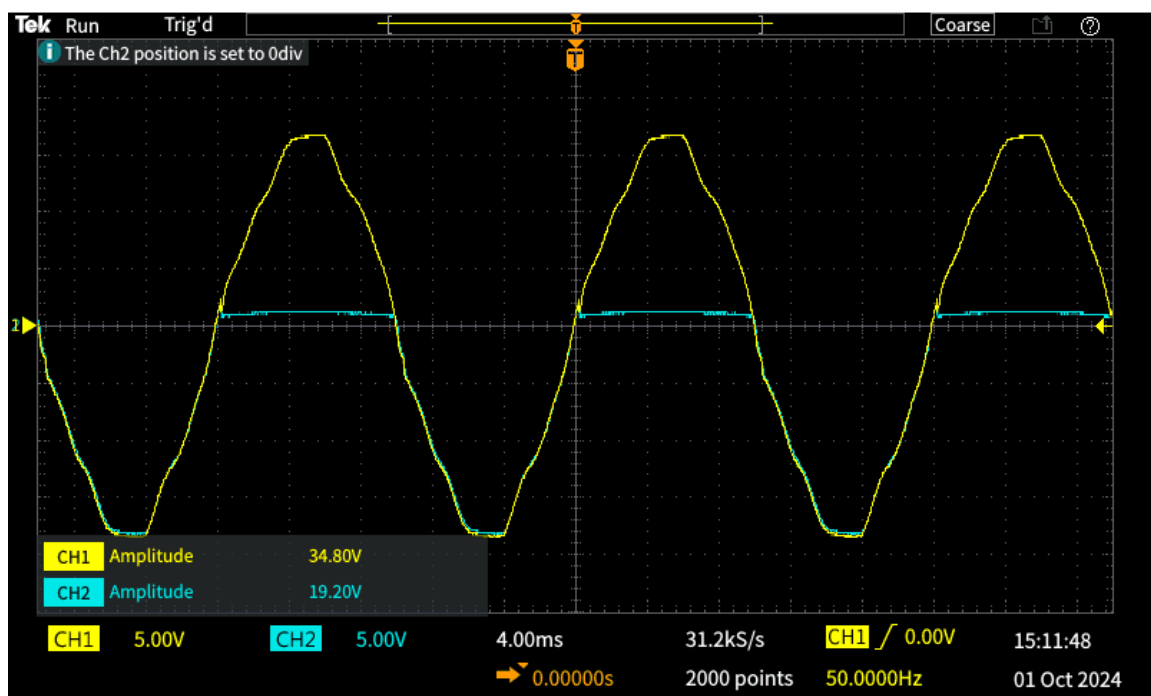
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Figure 2 Potentiometer 0% on



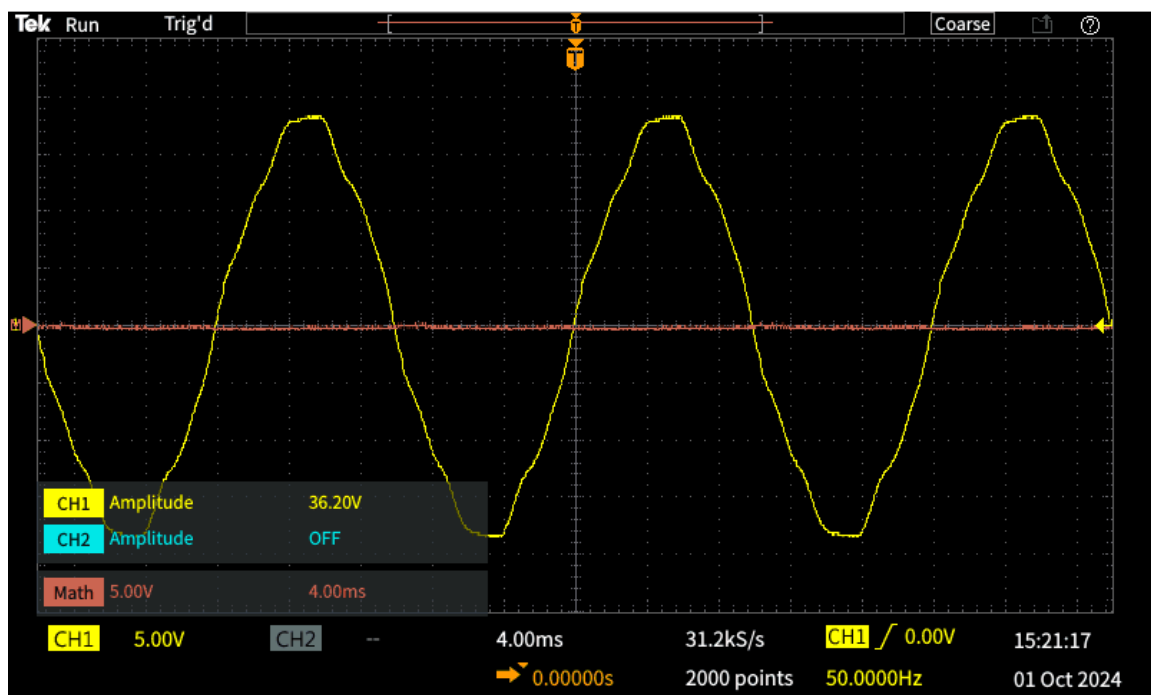
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Figure 3 Potentiometer 50% on



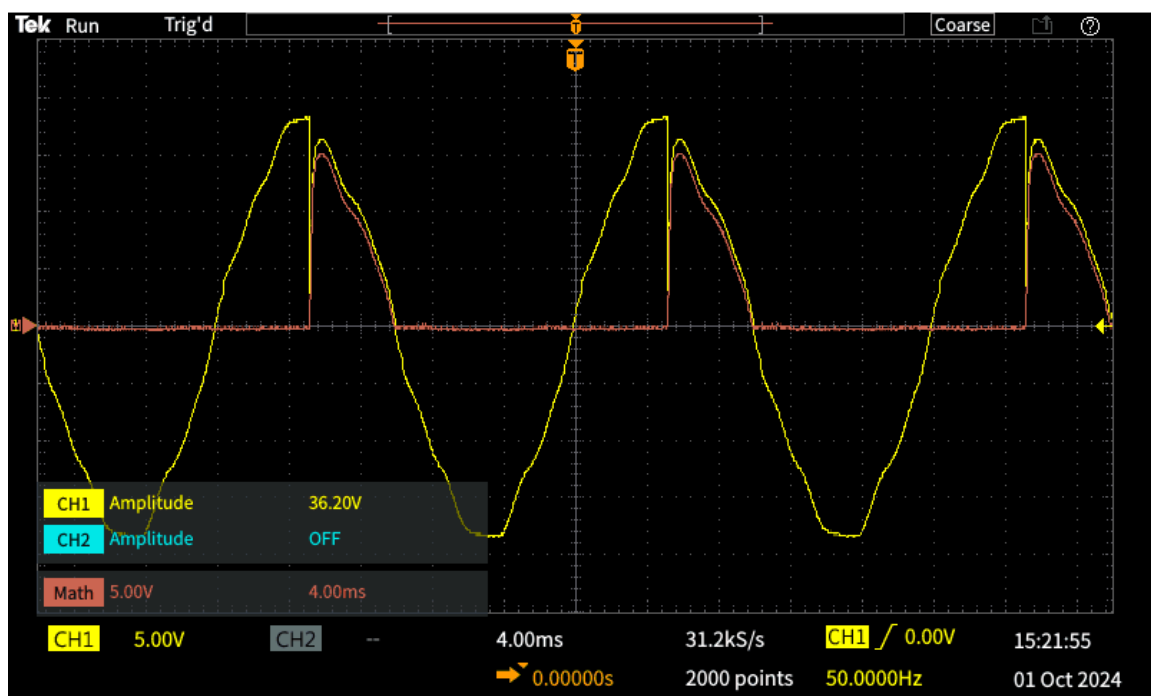
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Figure 4 Potentiometer 100% on



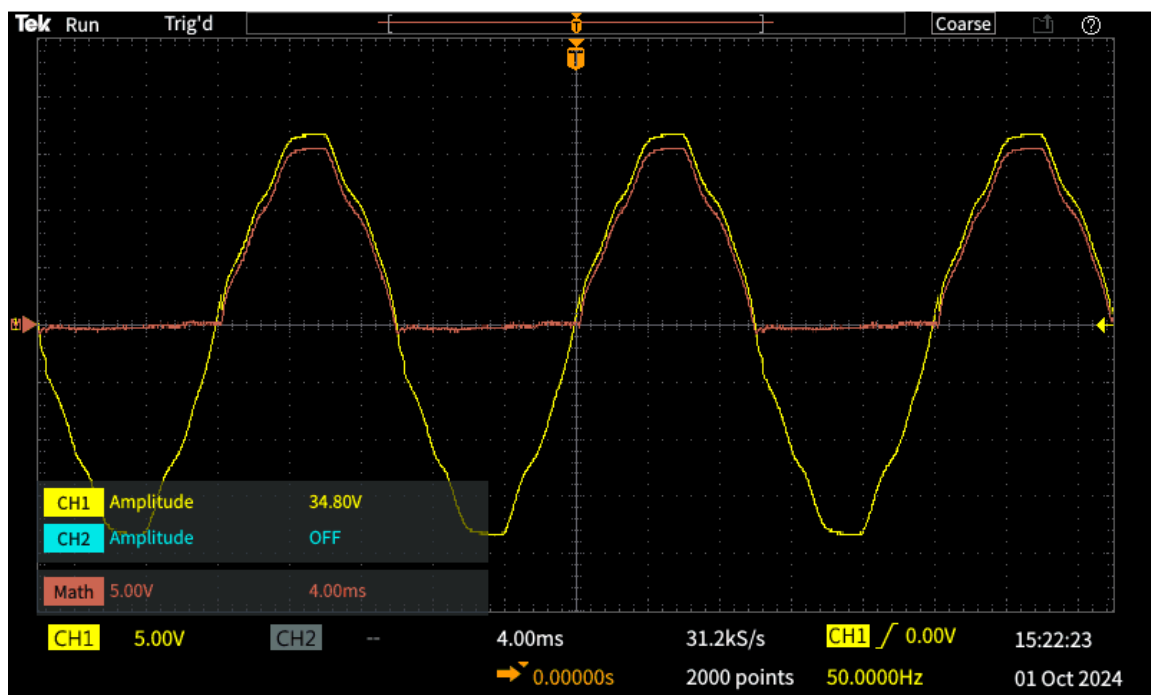
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Figure 5 Potentiometer 0% on



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Figure 6 Potentiometer 50% on



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Figure 7 Potentiometer 100% on

3.2 TRIAC

3.2.1 Measurements

The second circuit studied is the TRIAC in an AC circuit. A TRIAC is a three-terminal electrical component that conducts current in either direction when triggered. It can be characterised as a thyristor that functions in both directions. The TRIAC circuit is shown in Figure 8 below. Because of our low supply voltage, we remove the DIAC from the circuit. For this circuit, we vary the resistance of the potentiometer and observe how the voltage over the TRIAC changes

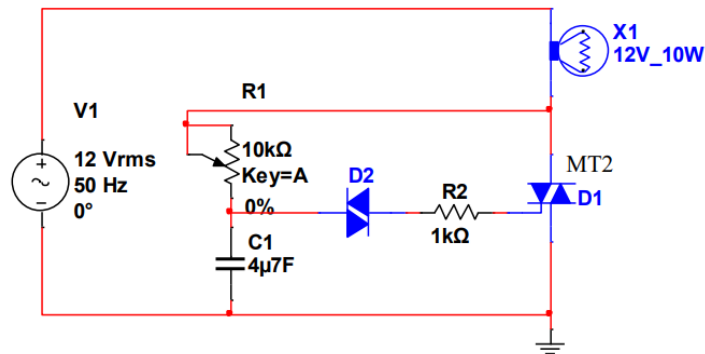
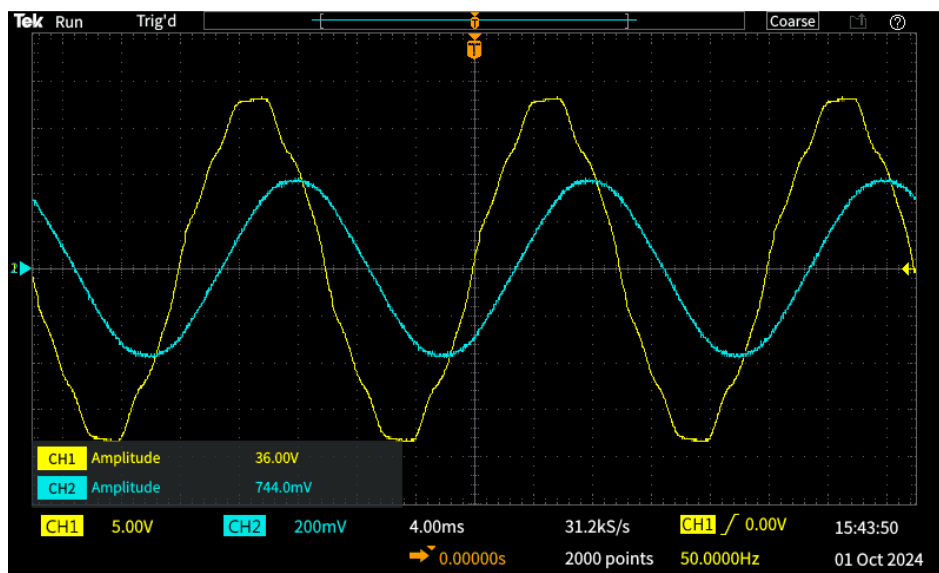


Figure 8: TRIAC in an AC circuit

3.2.2 Graphs

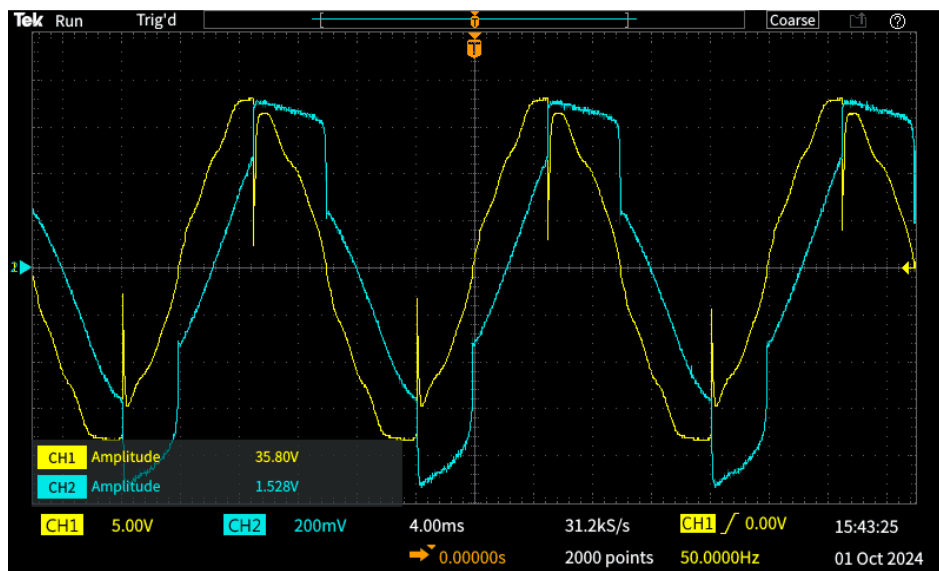
The six figures below show the voltage over the TRIAC with the potentiometer at 0%, 50%, and 100% respectively. First, we show the input signal and signal at the TRIAC. Then, we

subtract the input signal from the signal over the TRIAC to determine the signal over the load. Figures 9, 10, 11 therefore show the voltage over the load along with the input signal.



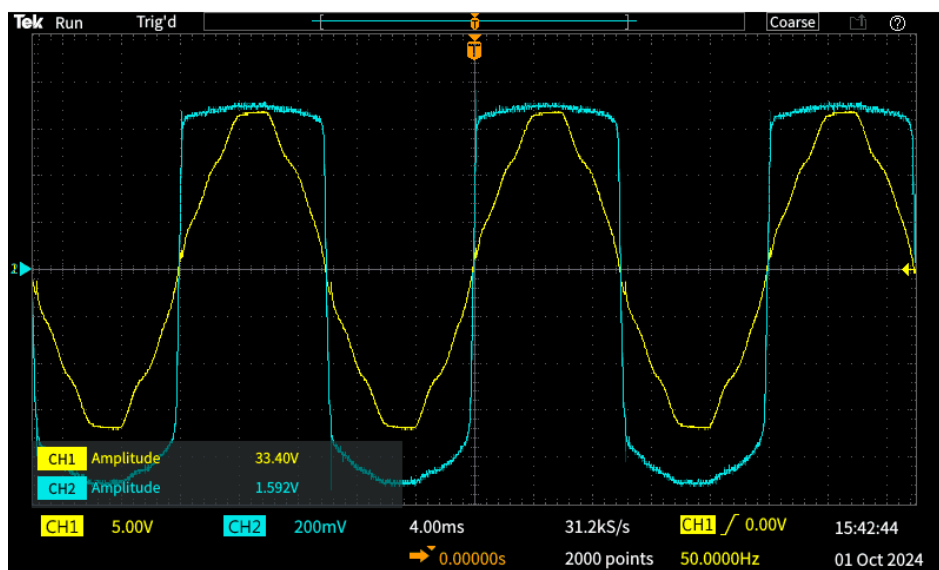
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Figure 9: Potentiometer 0% on



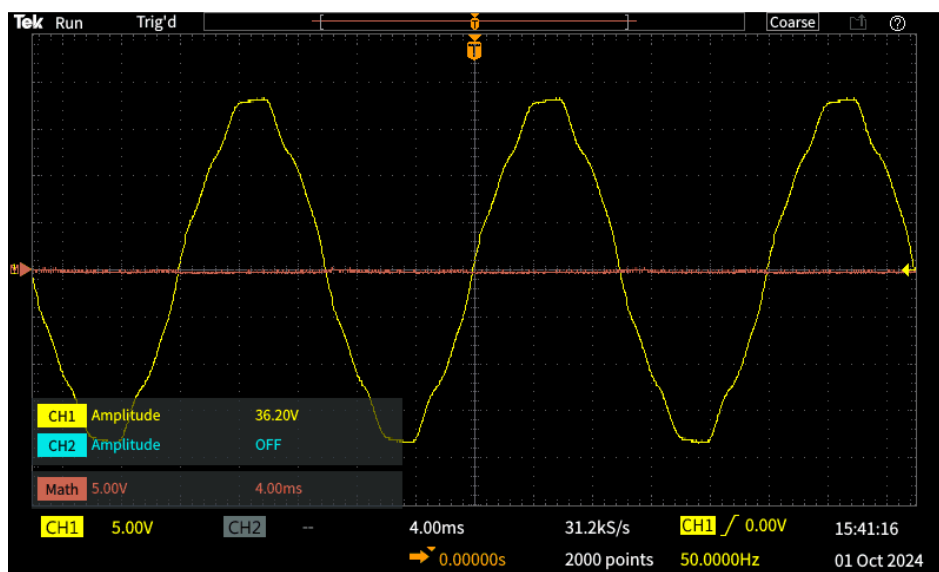
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Figure 10: Potentiometer 50% on



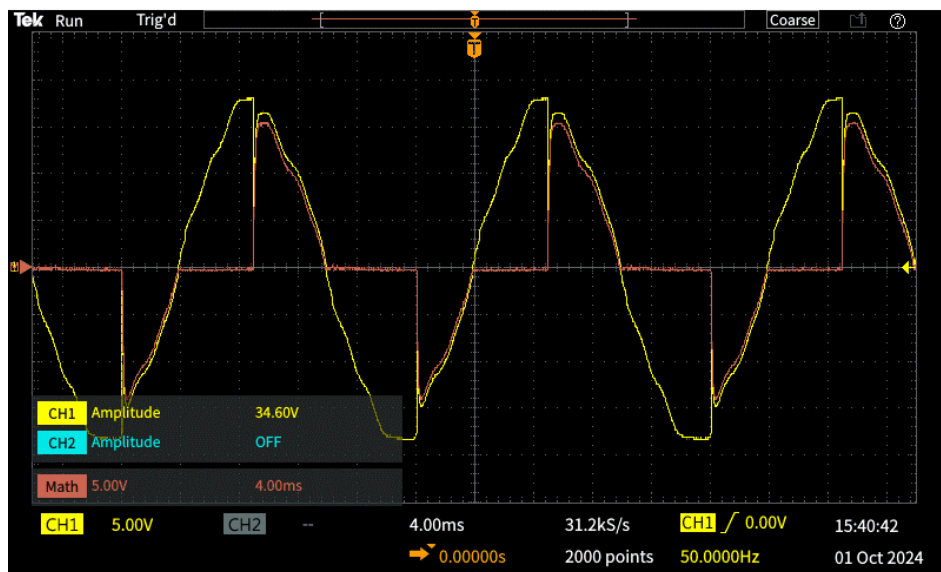
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Figure 11: Potentiometer 100% on



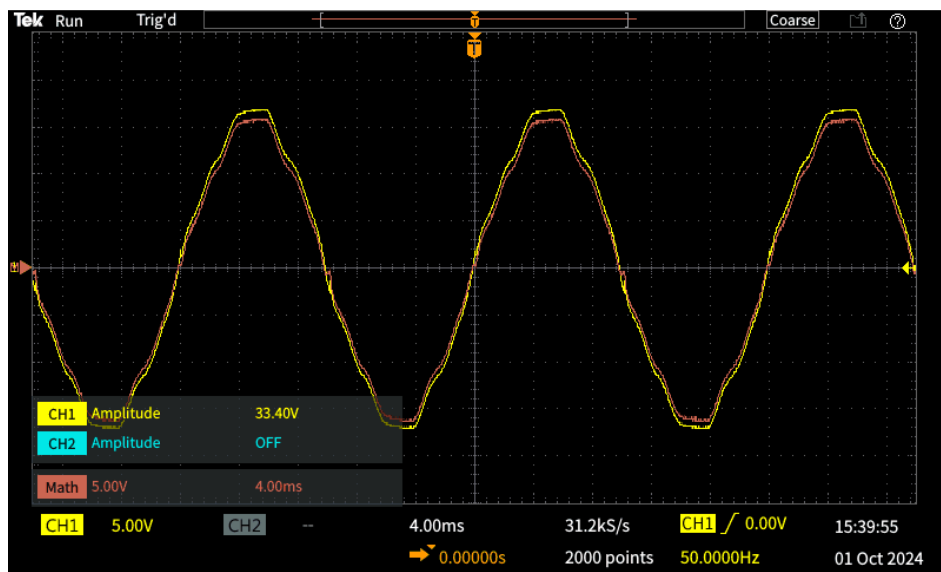
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Figure 12: Potentiometer 0% on



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Figure 13: Potentiometer 50% on



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Figure 14: Potentiometer 100% on

3.3 Conclusion

As we can see, the use of TRIAC in place of a thyristor means the circuit fires symmetrically. We can see this from the different waveforms for when the gate is fully on. In the thyristor AC circuit, the thyristor is only conducting during the positive excursions of the input signal and is off during the negative excursions. In the TRIAC circuit, the component is conducting in both

excursions as can be clearly seen in figures 11 and 14. In general, we also observe a smoother and more symmetrical firing as can be noted by comparing the two circuits when the potentiometer is 50% on. The phase behaviour of the two circuits, however, is similar.

4 RESULTS MOSFET AND IGBT

4.1 Measurements

The next circuits studied revolve around MOSFETs and IGBTs. IGBTs combine the advantages from MOSFETs and bipolar transistors into one device using an input MOS and an output bipolar transistor. The circuit studied below in Figure 15 is a switching circuit using an IGBT.

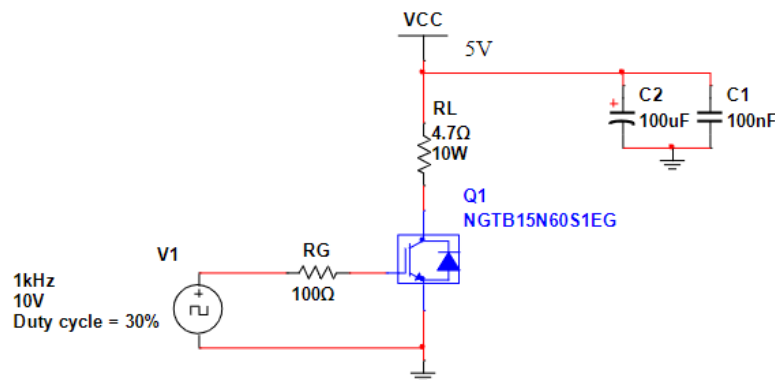
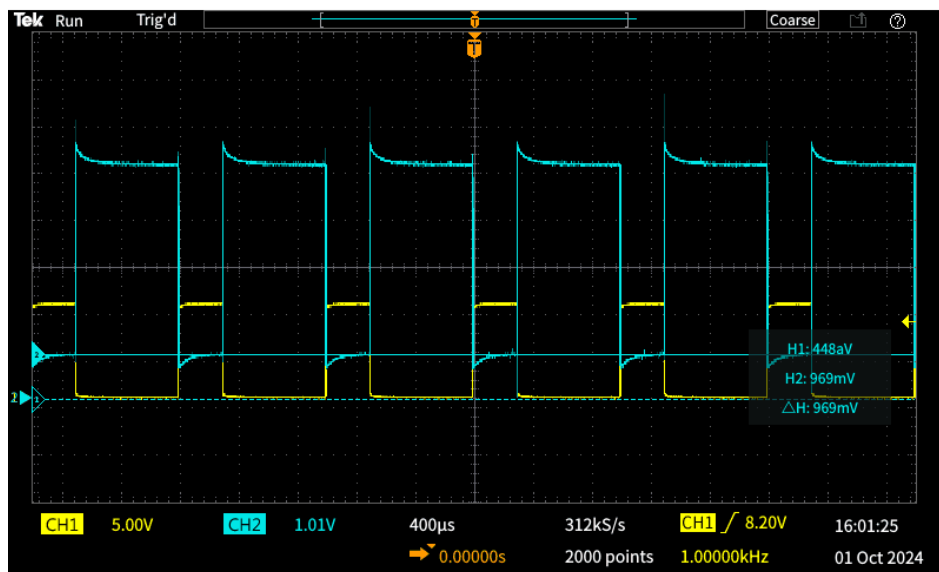


Figure 15: Switching a Resistive Load with an IGBT

4.2 Graphs

In the figure below, we show the waveform over the IGBT when it is brought into saturation. The NGTB15N60S1EG IGBT has a rated saturation voltage of $V_{CE,SAT} = 1.5V$ at $15A, 600V$. Given that we are operating at lower currents and voltages, we expect the measured saturation voltage to be lower. This is because the internal resistive components contribute less voltage drop at lower currents, which reduces the overall saturation voltage. The lower the current, the smaller the saturation voltage will be, though it will never be zero due to the inherent properties of the device. Figure 16 below shows the measured saturation voltage at approximately $970mV$ which indeed corresponds to our expectation and is in line with the conditions at which the IGBT was operated.



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Figure 16: IGBT Saturation Voltage

5 RESULTS CLAMPING CIRCUITS

5.1 Measurements

The final section of the lab focusses on clamping circuits. All the built circuits can be seen in figures 17-22 below. Each circuit exhibits different characteristics in its switching and clamping abilities. When an inductive load is switched on, energy will be stored inside its magnetic field. When switched off, current should keep on flowing through the inductor to ensure the inductor can remove its saved energy. If no current path is present when the transistor turns off, however, high voltage peaks will be present which can destroy the switching element.

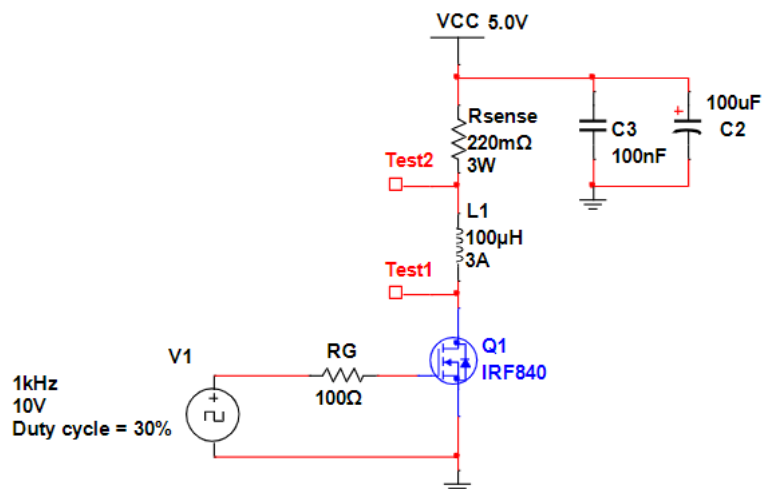


Figure 17: Switching an Inductive Load Without Clamping

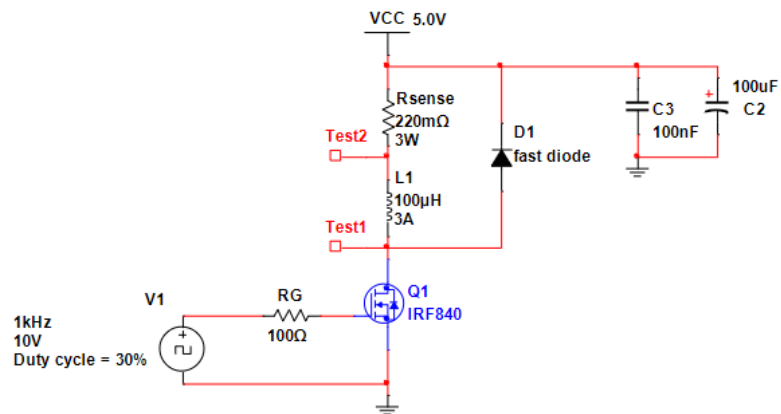


Figure 18: Switching an Inductive Load using only a Diode

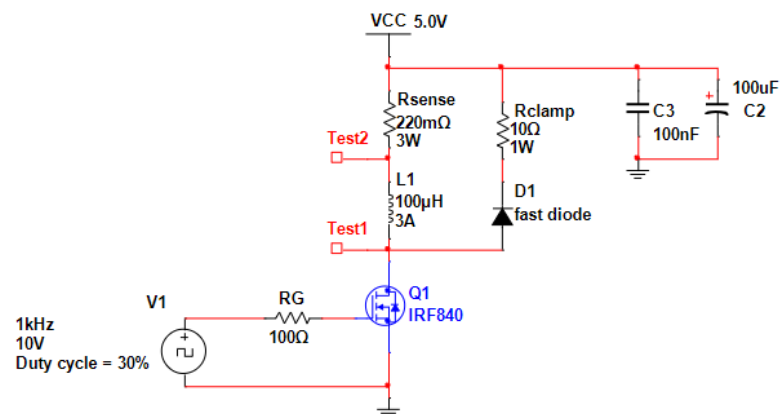


Figure 19: Switching an Inductive Load using a Diode and Resistor

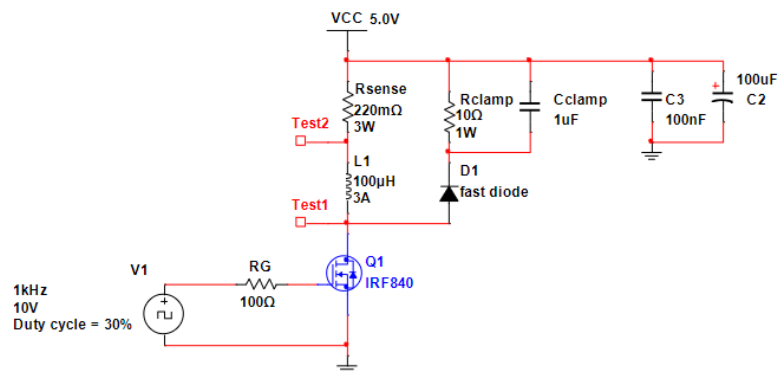


Figure 20: Switching an Inductive Load using a Diode, Resistor, and Capacitor

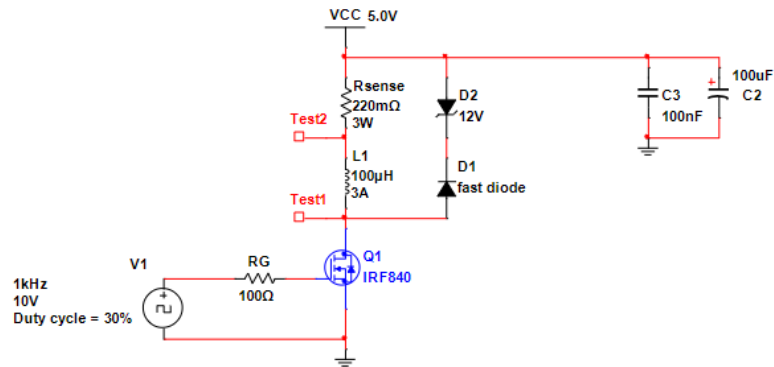


Figure 21: Switching an Inductive Load using a Diode and Zener Diode

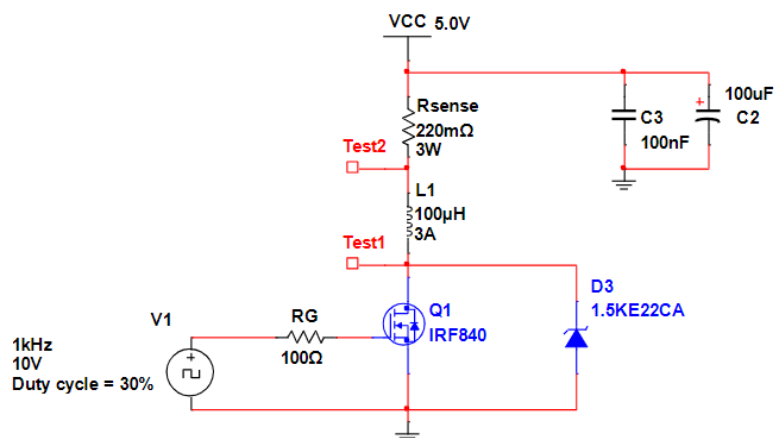


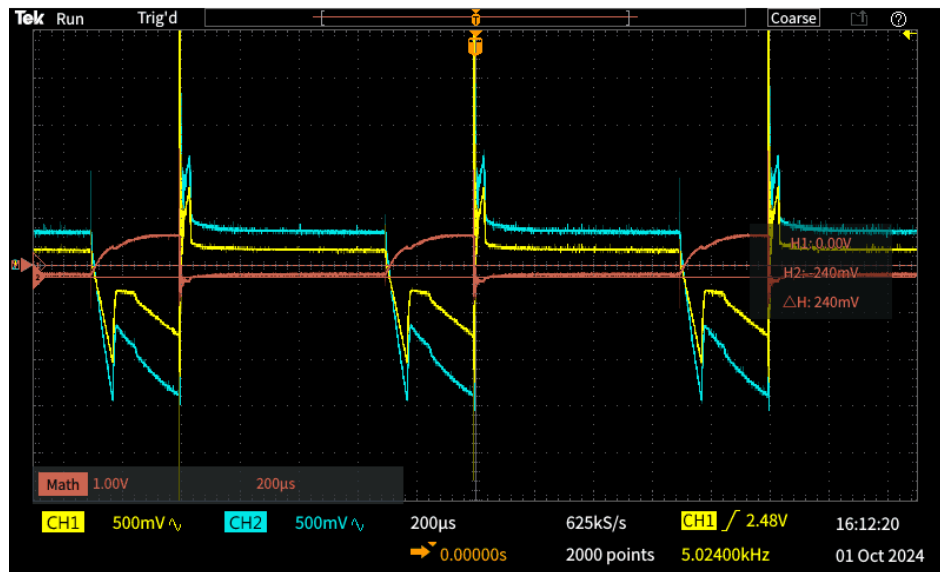
Figure 22: Switching an Inductive Load using a Transient Voltage Suppressor (TVS)

5.2 Graphs

Due to parasitic effects there are anomalies in our graphs as can be seen by irregular peaks and deviations from the fully expected behavior, but the essential effects can still be observed.

5.2.1 No clamping

The first circuit studied (figure 23) does not have any clamping mechanism in place.



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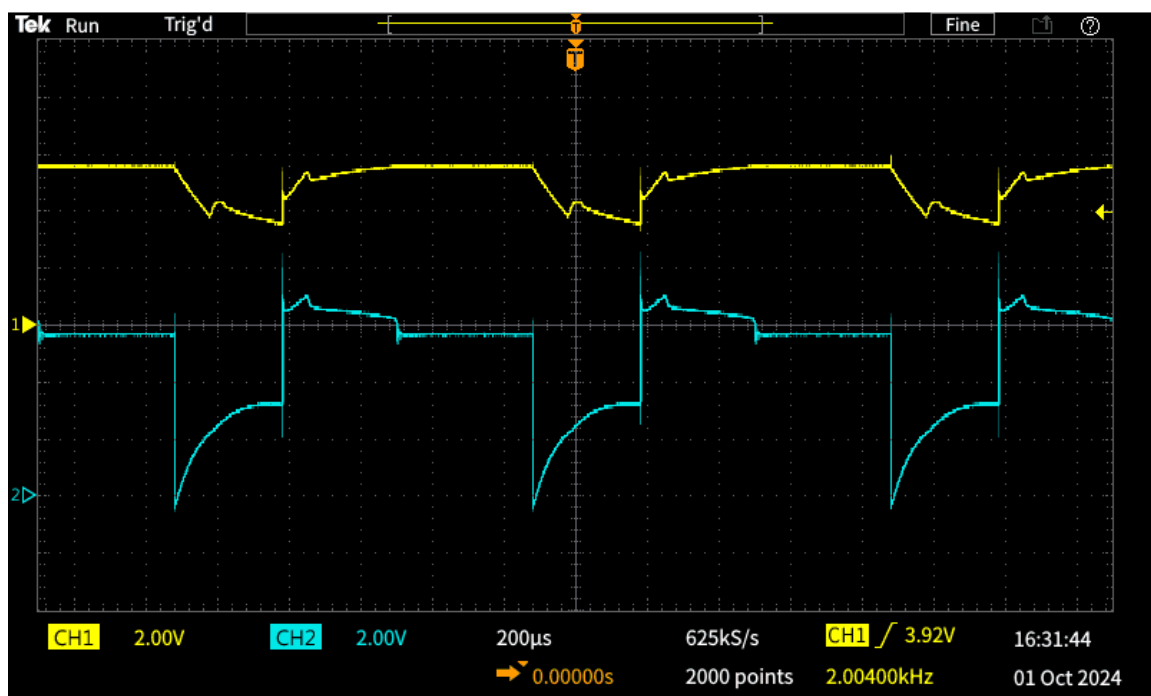
Figure 23: No clamping signals and current

The voltage peaks in this circuit are significantly more energetic than the ones we will see in subsequent circuits. This is easily explained by the lack of any clamping or suppression mechanism.

5.2.2 Only diode

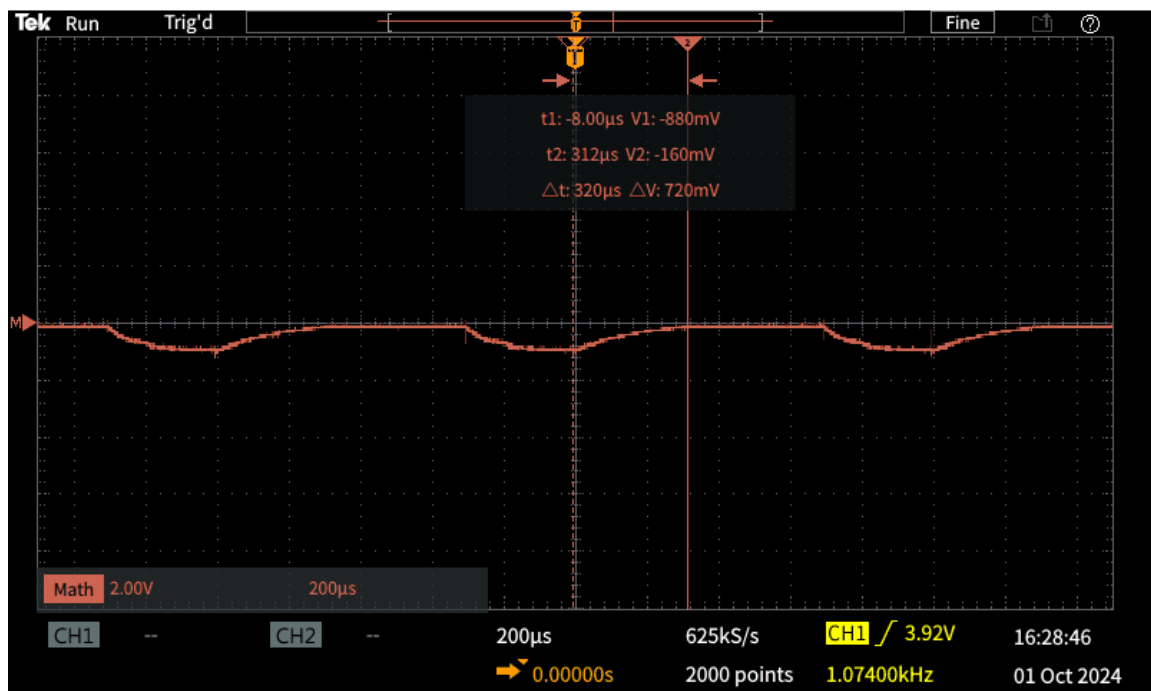
In the second circuit, a diode is used to make a closed current loop so that the power generated by the inductive load (back emf) can be dissipated. The diode and the load itself dissipate the energy. However, this method is not very efficient, as the voltage across the diode is only around 0.7V, which limits its ability to dissipate significant power.

In the next circuit, a power resistor is added to the loopback path. This increases the voltage across the path, allowing the resistor to dissipate the energy more effectively, resulting in a more efficient protection for the circuit.



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Figure 24 Diode Clamping Circuit Signals

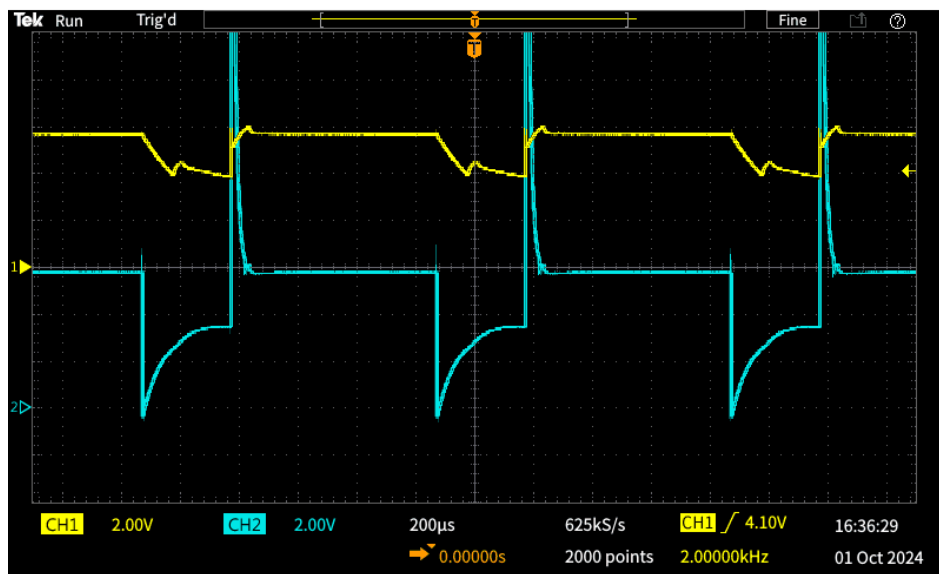


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Figure 25 Diode Clamping Circuit Current

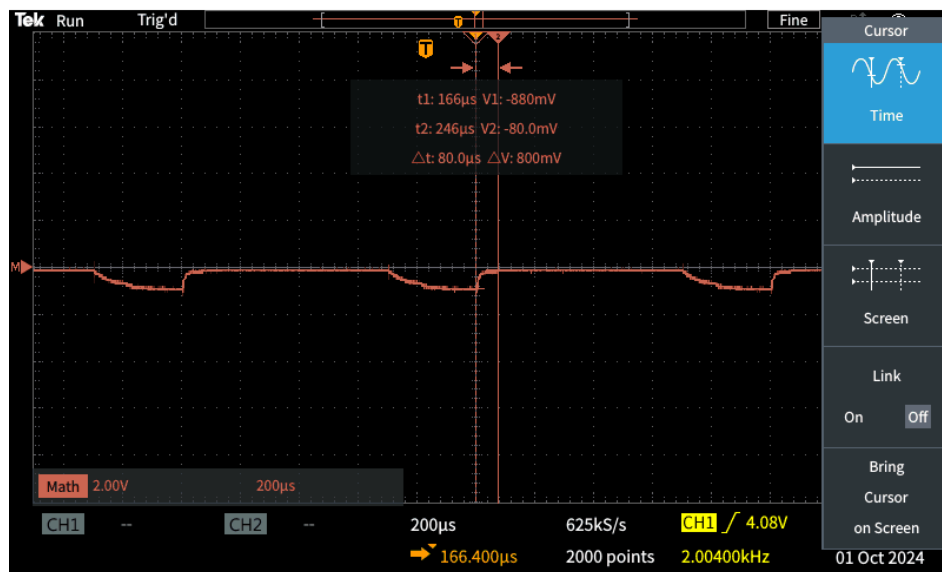
5.2.3 Diode + resistor

The next circuit studied (figure 19) is the one using a diode with a 10Ω power resistor. This scheme extends the one in figure 18 wherein the addition of the resistor provides a mechanism for faster power dissipation and current decay rather than having the current flow through the inductor again in a loop. This should therefore allow for a faster and smoother dissipation of energy in comparison to the previous implementation. The two figures below (26, 27) show the signals over the load. These signals are less energetic than the previous schemes as is expected and show a smoother and faster transient $80\mu\text{s}$ vs. $320\mu\text{s}$ in figure 27. The addition of the resistor simply accelerates power dissipation.



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Figure 26: Diode and Resistor Clamping Circuit Signals

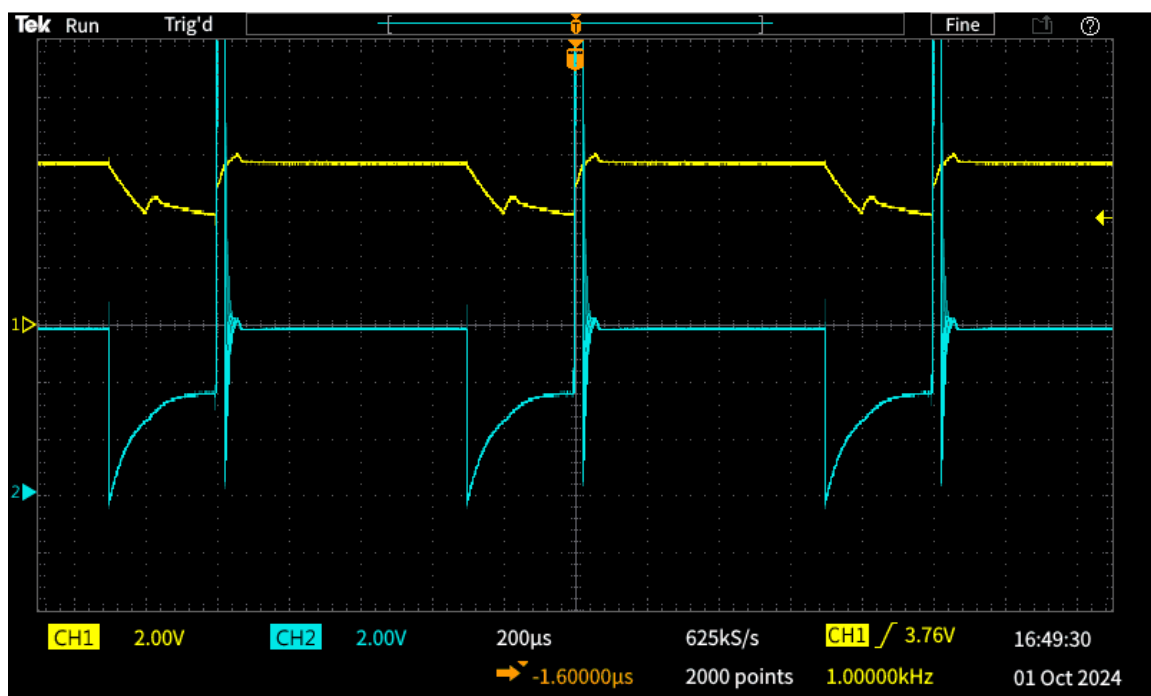


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Figure 27: Diode and Resistor Clamping Circuit Current

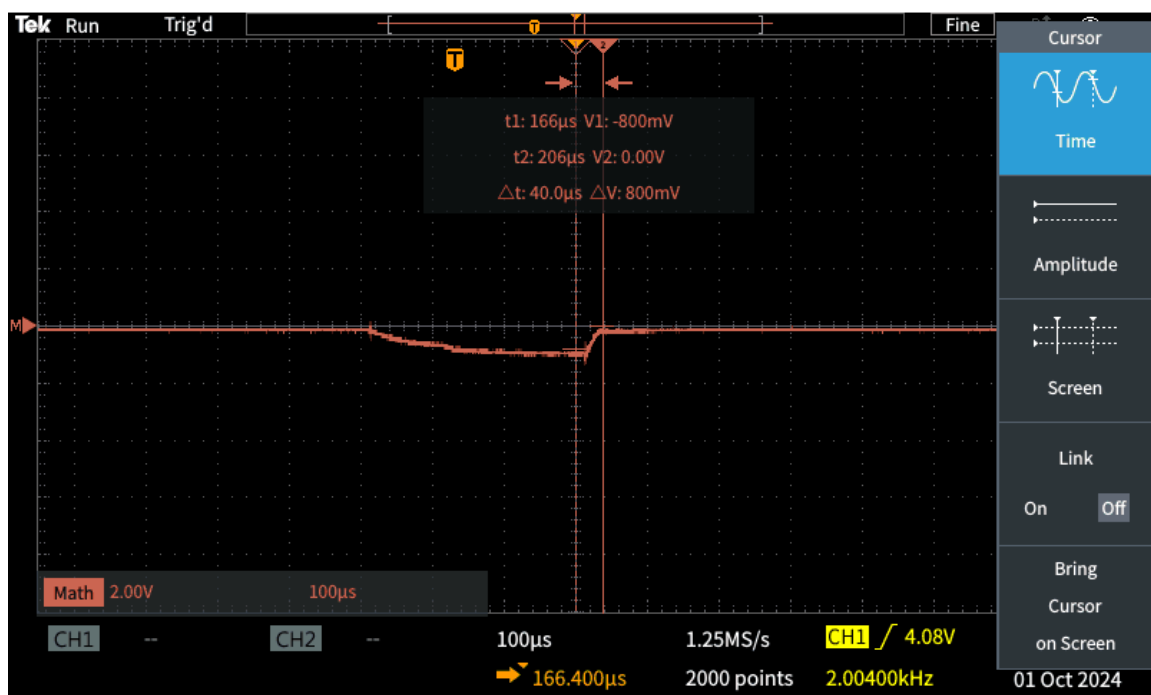
5.2.4 Diode + resistor + capacitor

In this circuit a capacitor is added to the closed current loop. This would further improve the circuit because it can rapidly charge during the voltage spike and then release its energy slowly. Because the capacitor can respond fast to the voltage spike the time the current will drop faster than all the previous circuits (40µs). The disadvantage of this circuit is that the capacitor will also introduce an oscillation with the inductor. This can be suppressed by choosing the right capacitor value. Choosing the right capacitor can also make the circuit faster. In the figures below, we show the behaviour of the circuit for capacitors of capacitance 1µF, 100nF, and 1nF.



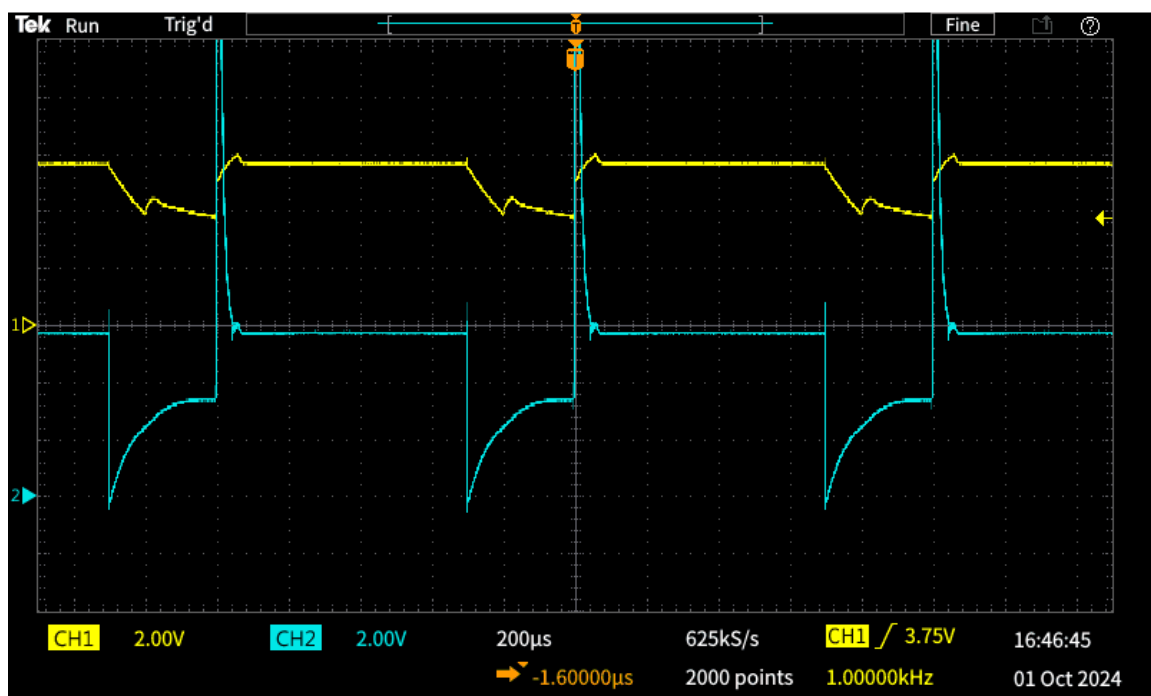
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Figure 28 Diode + Resistor + Capacitor (1µF) Clamping Circuit Signals



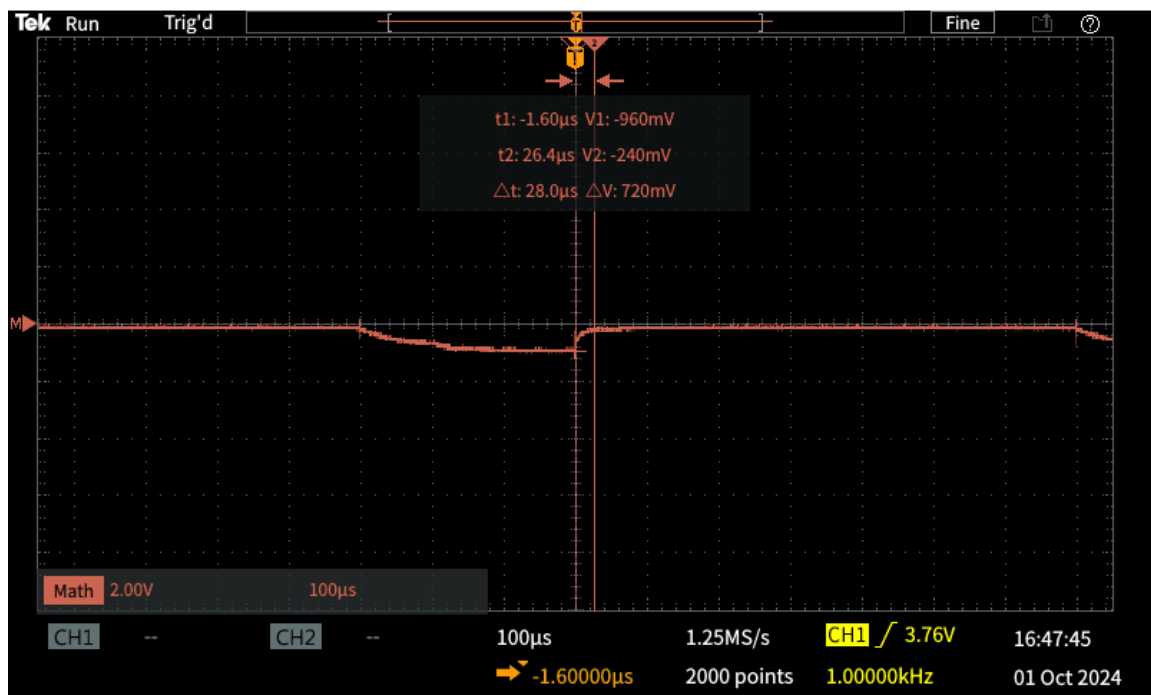
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Figure 29 Diode + Resistor + Capacitor (1µF) Clamping Circuit Current



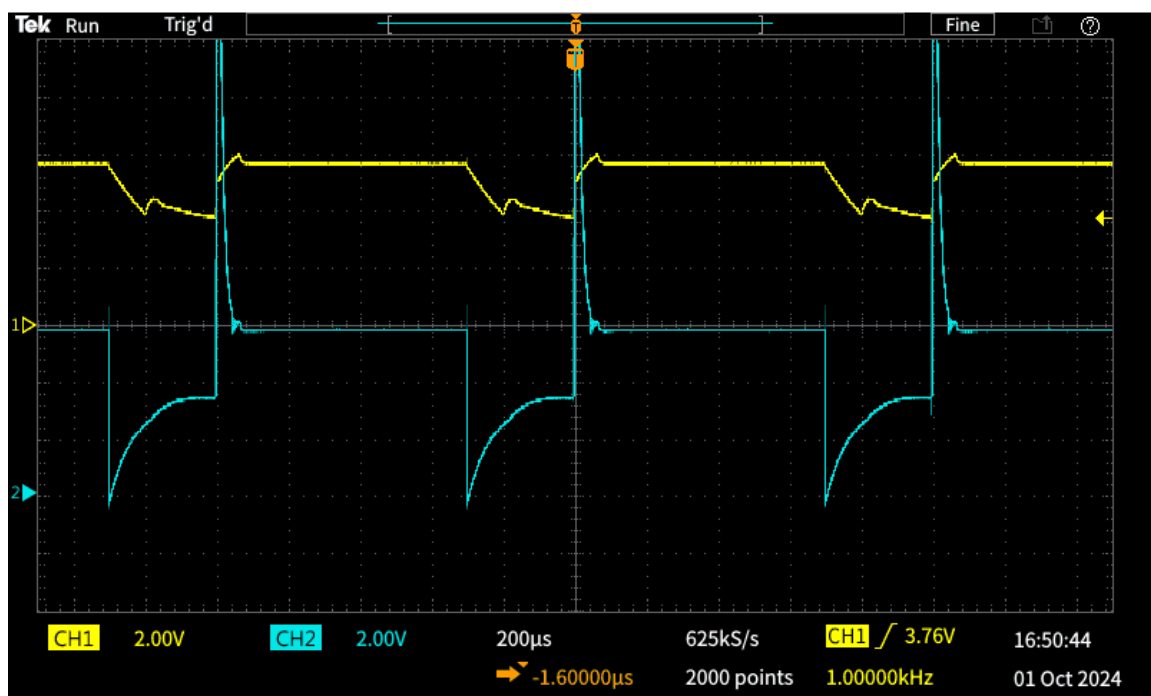
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Figure 30 Diode + Resistor + Capacitor (100nF) Clamping Circuit Signals



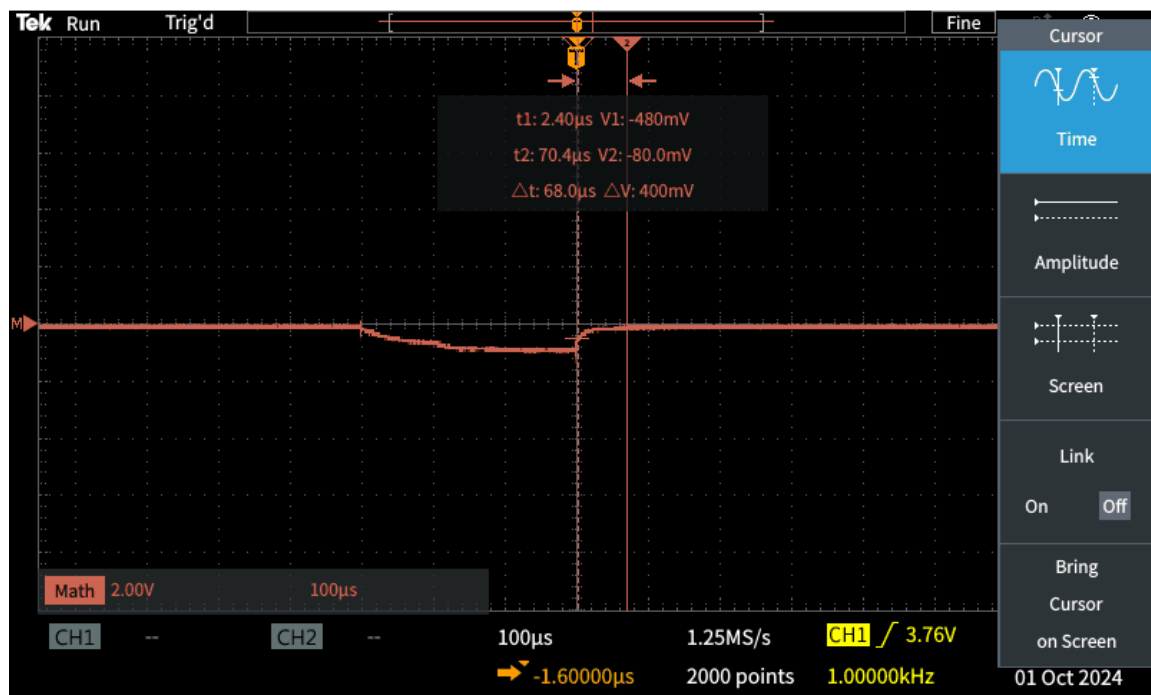
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Figure 31 Diode + Resistor + Capacitor (100nF) Clamping Circuit Current



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Figure 32 Diode + Resistor + Capacitor (1nF) Clamping Circuit Signals

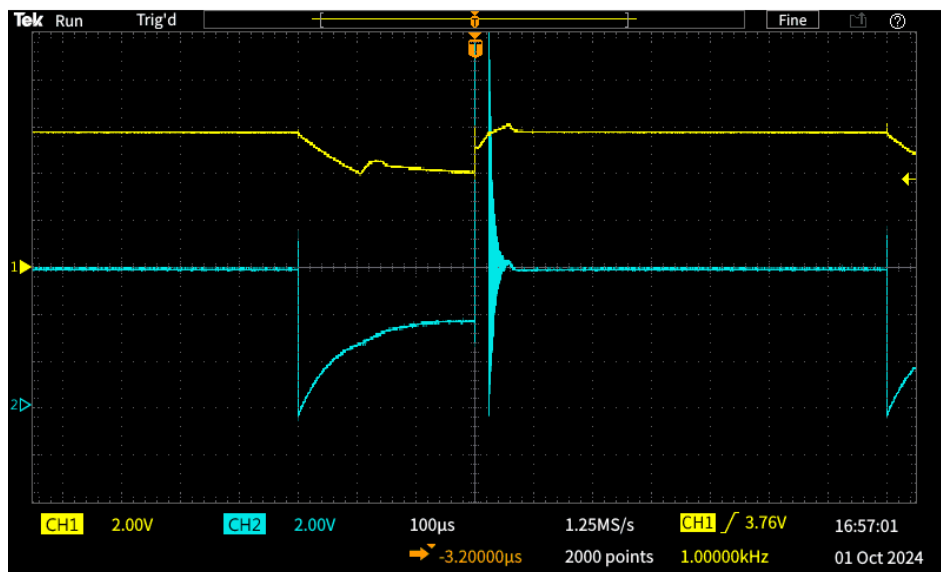


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Figure 33 Diode + Resistor + Capacitor (1nF) Clamping Circuit Current

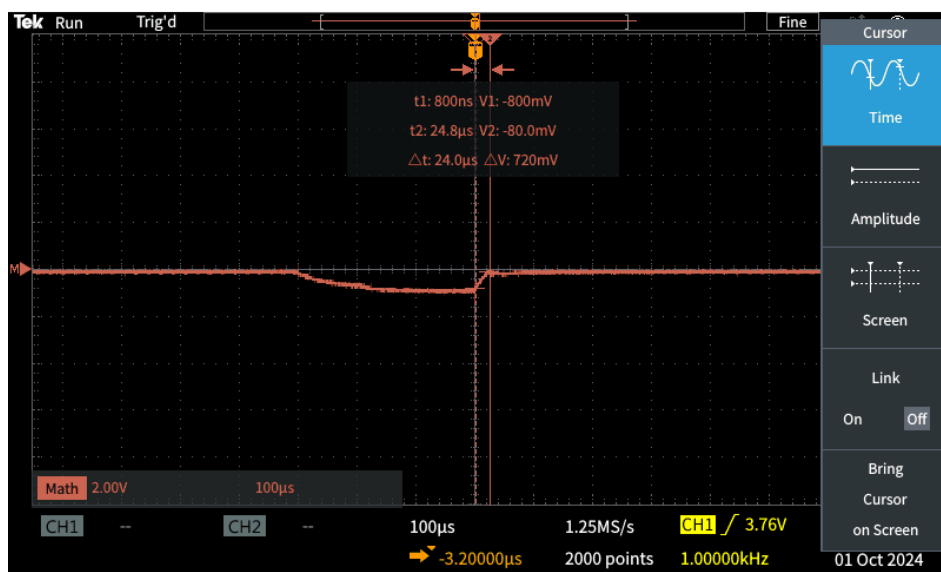
5.2.5 Diode + zener

The next circuit studied is (figure 21) which makes use of a diode and Zener diode in series. Here, the Zener diode is used to clamp the peak voltage and to dissipate the energy out of the inductor. We show the behaviour of this circuit in figures 34 and 35 below. The addition of the Zener allows the voltage spike to rise higher in exchange for a faster current decay $24\mu s$. Because of the higher voltage over the Zener a higher power can be dissipated. Essentially, here we are trading off voltage spike height with speed of dissipation. Another phenomenon we observe in this circuit is ringing. As we can see, after switching, the voltage signal over the transistor oscillates. The oscillations arise due to the interaction between the inductance of the load and the parasitic capacitances in the circuit. The parasitic capacitances in the circuit can form an LC circuit with the inductance of the load. Once the initial voltage spike happens and energy is stored in the parasitic capacitance, this stored energy can oscillate between the inductor and the capacitance, forming a resonant tank circuit which is what we observe in figure 34. Given the components used, it is impossible to get rid of these oscillations.



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Figure 34: Diode and Zener Diode Clamping Circuit Signals



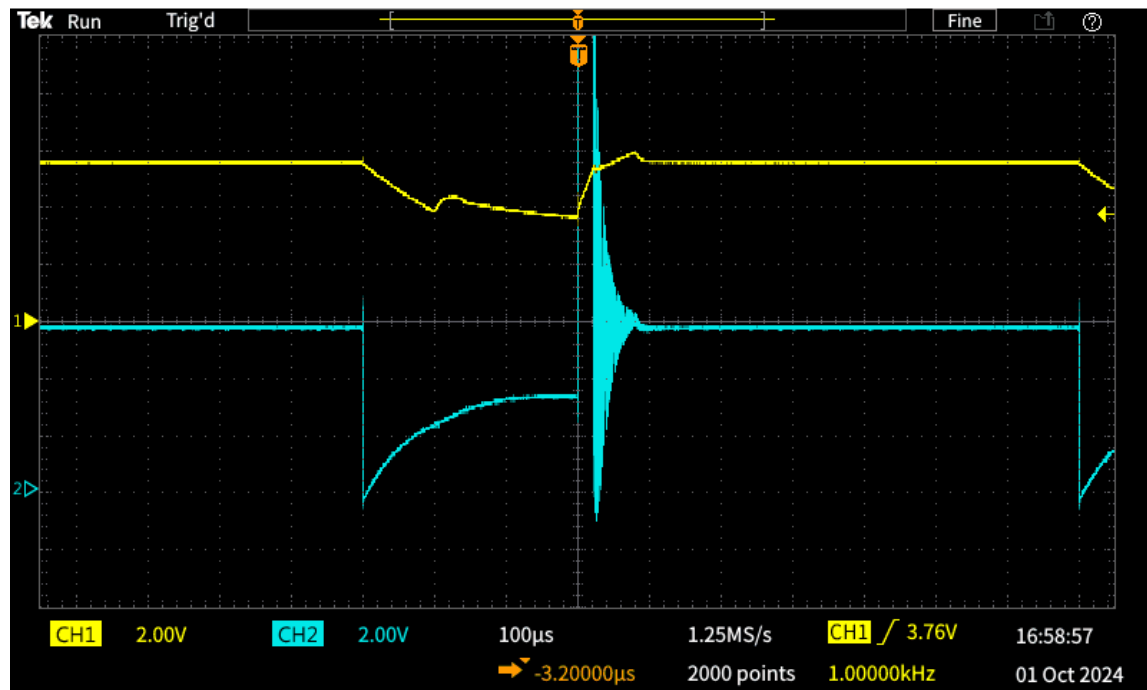
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Figure 35: Diode and Zener Diode Clamping Circuit Current

5.2.6 Transient suppressor

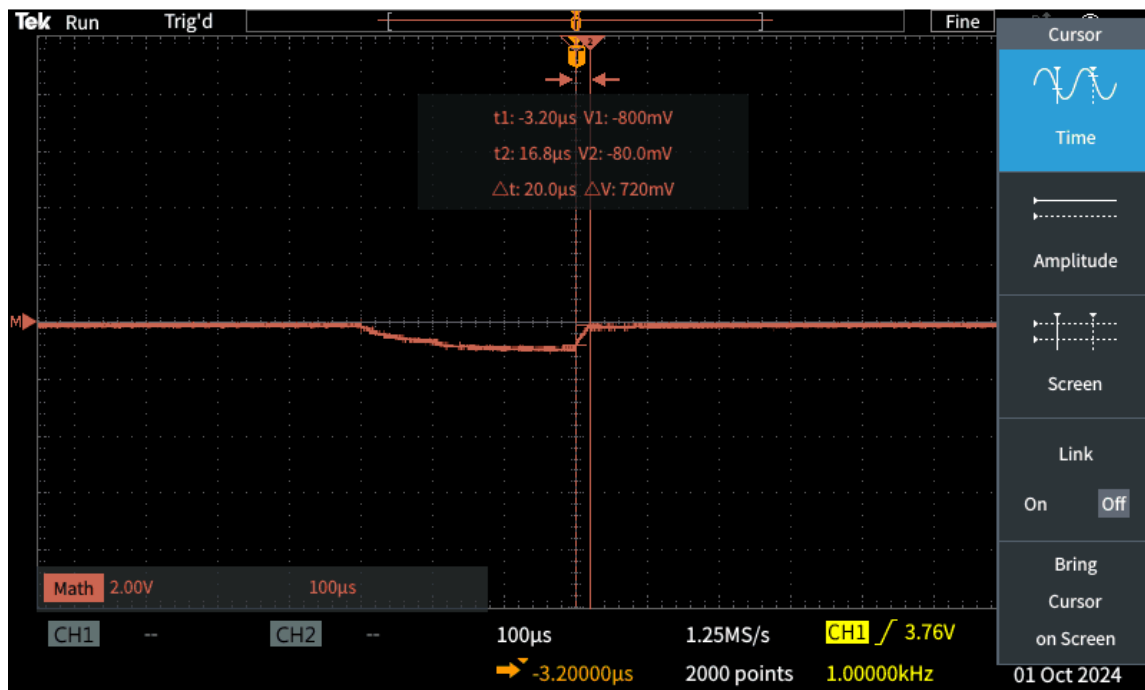
In this circuit a 22V TVS diode is placed directly over the FET. This will ensure that the FET will not break. The TVS will dissipate all the power until it has a safe voltage of 22 Volts. The principle is the same as the circuit with the Zener diode, but the TVS is designed for big peak powers. This method will also give the fastest current drop time of 20µs. The downside here is

that when it's below the breakdown voltage of 22 there is no closed current loop so there will be oscillations.



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Figure 36 TVS Clamping Circuit Signals



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Figure 37 TVS Clamping Circuit Current

5.3 Conclusion

In this final section, we examined different types of clamping mechanisms when dealing with inductive loads. As we can see from the circuits studied above, effective clamping methods are complex and often involve trade-offs between time, oscillatory behaviour, and energy spikes. Each particular circuit has its benefits and drawbacks and the ideal circuit depends on its application.

6 GENERAL CONCLUSION

Through this lab, we gained a better understanding of electronic components used in switching and clamping circuits. The main difference between the thyristor and TRIAC is that unlike the thyristor which is a unidirectional device, the TRIAC is bidirectional which allows firing in both directions and therefore more symmetrical behaviour. IGBTs combine the advantages from MOSFETs and bipolar transistors into one device. It is easy to drive IGBTs given the high input impedance at the gate which means very little current is required to control the device. Since the gate is voltage-driven rather than current-driven, the power required to switch the IGBT is relatively low, reducing the complexity and power consumption of the driver circuit. The chosen clamping circuit will determine various properties of your circuit but primarily current decay time

and power dissipation time, the presence of oscillations during switching, and the magnitude of voltage spikes during switching which allows us to determine which clamping mechanism is most appropriate based on the application of the circuit.